IP ADDRESSING AND SUBNETTING

1.List the IP address classes (A, B, C, D, E) and give one real-world example of a device or service that might use each class. Write your answers in a table format.

ANS:

IP Address Classes and Real-World Examples

IP Class	**IP Address Range**	**Purpose**	**Real-World Example**
Class A	1.0.0.0 – 126.255.255.255	Large networks	A large organization's internal network.
Class B	128.0.0.0 – 191.255.255.255	Medium-sized networks	A university campus network.
Class C	192.0.0.0 – 223.255.255.255	Small networks	A home router using 192.168.1.10 for a laptop (private address).
Class D	224.0.0.0 – 239.255.255.255	Multicasting	A multicast stream sent to multiple devices on a network.
Class E	240.0.0.0 – 255.255.255.255	Reserved/experimental	Network research and experimental purposes.

**2.Given the IP address 192.168.10.45 with subnet mask 255.255.255.0, calculate the network address, broadcast address, and the range of valid host addresses.

Hint: Show your working steps for each calculation.**

ANS:

Given Information

- IP Address: 192.168.10.45
- Subnet Mask: 255.255.255.0

Step 1: Calculate the Network Address

The network address is calculated using a bitwise AND operation between the IP address and subnet mask.

IP Address: 192.168.10.45

Subnet Mask: 255.255.255.0

Network Address:192.168.10.0

Answer: Network address = 192.168.10.0

Step 2: Calculate the Broadcast Address

Since the subnet mask is /24, the last 8 bits are reserved for host addresses. Set all host bits to 1.

Network Address: 192.168.10.0

Broadcast Address:192.168.10.255

Answer: Broadcast address = 192.168.10.255

Step 3: Calculate the Valid Host Address Range

- First valid host = Network address + 1
- Last valid host = Broadcast address − 1

First valid host: 192.168.10.1

Last valid host: 192.168.10.254

Answer: Valid host range = 192.168.10.1 – 192.168.10.254

Final Answer

Item	**Result**
IP Address	192.168.10.45
Subnet Mask	255.255.255.0 (/24)
Network Address	192.168.10.0
Broadcast Address	192.168.10.255
Valid Host Range	192.168.10.1 – 192.168.10.254
Total Valid Host Addresses	254

3.Create a subnetting plan for a small food delivery startup like Zomato that needs 4 separate subnets: one for Delivery Partners, one for Restaurants, one for Customers, and one for Admins. Assign a unique subnet (with network address and subnet mask) for each group using the 10.0.0.0/24 network.

ANS:

Subnetting Plan for a Food Delivery Startup

The startup uses the network 10.0.0.0/24 and divides it into 4 separate subnets for Delivery Partners, Restaurants, Customers, and Admins.

To create 4 equal subnets, we borrow 2 host bits:

- Original network: /24
- New subnet prefix: /26
- Subnet mask: 255.255.255.192
- Each subnet has 64 IP addresses, with 62 valid host addresses.

Subnetting Table

Group	Network Address	Subnet Mask	Valid Host Range	Broadcast Address
Delivery Partners	10.0.0.0/26	255.255.255.192	10.0.0.1 – 10.0.0.62	10.0.0.63
Restaurants	10.0.0.64/26	255.255.255.192	10.0.0.65 – 10.0.0.126	10.0.0.127
Customers	10.0.0.128/26	255.255.255.192	10.0.0.129 – 10.0.0.190	10.0.0.191
Admins	10.0.0.192/26	255.255.255.192	10.0.0.193 – 10.0.0.254	10.0.0.255

Calculation

1. The original /24 network contains 256 IP addresses.
2. We need 4 subnets, so we borrow 2 bits because \(2^2 = 4\).
3. The new prefix is /26, giving \(2^{32-26} = 64\) addresses per subnet.
4. Valid hosts per subnet = \(64 - 2 = 62\).

**4.You are given the following scenario: A college WiFi network uses the IP range 172.16.0.0/16. The admin wants to create at least 20 subnets for different buildings. Calculate the new subnet mask and the number of hosts available per subnet.

Constraint: Show your calculation steps and final answers.**

ANS:

Subnetting Calculation for a College Wi-Fi Network

Given:

- Network address: 172.16.0.0/16
- Required subnets: At least 20

Step 1: Calculate the Bits to Borrow

Formula:

$$ 2^n \geq \text{Required subnets} $$

$$ 2^n \geq 20 $$

Since $2^4 = 16$ (not enough) and $2^5 = 32$,

Bits borrowed = 5

Step 2: Calculate the New Subnet Mask

Original prefix = /16

New prefix = $16 + 5 = /21$

The new subnet mask is:

255.255.248.0

Step 3: Calculate Hosts per Subnet

Remaining host bits:

$$ 32 - 21 = 11 $$

Total IP addresses per subnet:

$$ 2^{11} = 2048 $$

Usable host addresses:

$$ 2048 - 2 = 2046 $$

We subtract 2 for the network address and broadcast address.

Final Answers

Item	**Answer**
Original Network	172.16.0.0/16
Required Subnets	At least 20
Bits Borrowed	5
Total Subnets Created	32
New CIDR Prefix	/21
New Subnet Mask	255.255.248.0
Total IP Addresses per Subnet	2048
Usable Hosts per Subnet	2046

5.A friend sends you a message asking why their device with IP 192.168.1.300 is not connecting to the network. Write a short explanation identifying the error and what a valid IP address in that subnet should look like.

ANS:

The IP address 192.168.1.300 is invalid because each octet in an IPv4 address must be between 0 and 255. Here, 300 exceeds the maximum value.

A valid IP address in the same subnet could be 192.168.1.100, assuming it is not already assigned to another device and is within the network's allowed range.